

FREE GROUPS

These are problems about free groups, their automorphisms and related issues. See also problems (O1), (O7), (O8), (O9), (O10) in the [OUTSTANDING PROBLEMS](#) section.

(F1)* (a) Is there an algorithm for deciding if a given automorphism of a free group has a non-trivial fixed point ? [Background](#)

(b) Is there an algorithm for deciding if a given finitely generated subgroup of a free group is the fixed point group of some automorphism ? [Background](#)

*(F2) (H.Bass) Does the automorphism group of a free group satisfy the "Tits alternative" ? [Background](#)

*(F3) (V.Shpilrain) If an endomorphism ϕ of a free group F of finite rank takes every primitive element to another primitive, is ϕ an automorphism? [Background](#)

*(F4) Denote by $\text{Orb}_{\phi}(u)$ the orbit of an element u of the free group F_n under the action of an automorphism ϕ . That is, $\text{Orb}_{\phi}(u) = \{v \in F_n, v = \phi^m(u) \text{ for some } m \geq 0\}$. If an orbit like that is finite, how many elements can it possibly have if u runs through the whole group F_n , and ϕ runs through the whole group $\text{Aut}(F_n)$? [Background](#)

(F5) (H.Bass) Is the automorphism group of a free group "rigid", i.e., does it have only finitely many irreducible complex representations in every dimension? [Background](#)

(F6) The conjugacy problem for the automorphism group of a free group of finite rank. [Background](#)

*(F7) (V.Shpilrain) Denote by $\text{Epi}(n, k)$ the set of all homomorphisms from a free group F_n onto a free group F_k ; $n, k \geq 2$. Are there 2 elements $g_1, g_2 \in F_n$ with the following property: whenever $\phi(g_i) = \psi(g_i)$, $i = 1, 2$, for some homomorphisms $\phi, \psi \in \text{Epi}(n, k)$, it follows that $\phi = \psi$? (In other words, every homomorphism from $\text{Epi}(n, k)$ is completely determined by its values on just 2 elements). [Background](#)

(F8) (W.Dicks, E.Ventura) Let f be an endomorphism of a free group F_n , S a subgroup of F_n having finite rank. Is it true that $\text{rank}(\text{Fix}(f) \cap S) \leq \text{rank}(S)$? [Background](#)

(F9) (A.I.Kostrikin) Let F be a free group of rank 2 generated by x, y . Is the commutator $[x, y, y, y, y, y]$ a product of fifth powers in F ? (If not, then the Burnside group $B(2, 5)$ is infinite).

*(F10) (A.I.Mal'cev) Can one describe the commutator subgroup of a free group by a first order formula in the group theory language ? [Background](#)

(F11) (G.Bergman) Let S be a subgroup of a free group F , R a retract of F . Is it true that the intersection of R and S is a retract of S ? [Background](#)

*(F12) (G.Baumslag) Let $F = F_n$ be a free group generated by $\{x_1, \dots, x_n\}$, and let F^Q be the free Q -group, i.e., the free object of rank n in the category of uniquely divisible groups. Consider a canonical map $x_i \mapsto (1 + x_i)$ from F^Q into the formal power series ring Δ_Q with coefficients in Q . It is known that this map induces a homomorphism $\lambda : F^Q \rightarrow \Delta_Q$ (Magnus homomorphism). Is λ injective? Or, equivalently, is the group F^Q residually torsion-free nilpotent? [Background](#)

(F13) (I.Kapovich) Is the group F^Q in the previous problem linear? [Background](#)

(F14) Let F be a non-cyclic free group of finite rank, and G a finitely generated residually finite group. Is G isomorphic to F if it has the same set of finite homomorphic images as F does? [Background](#)

(F15) (V.Shpilrain) Let F be a non-cyclic free group, and R a non-cyclic subgroup of F . Suppose the commutator subgroup $[R, R]$ is a normal subgroup of F . Is R necessarily a normal subgroup of F ?

[Background](#)

*(F16) (V.Remeslennikov) Let R be the normal closure of an element r in a free group F with the natural length function, and suppose that s is an element of minimal length in R . Is it true that s is conjugate to one of the following elements: r , r^{-1} , $[r, f]$, or $[r^{-1}, f]$ for some element f ?

[Background](#)

*(F17) (V.Shpilrain) Let u be an element of a free group F . We call u a *strong test element* if, whenever $f(u) \neq 1$ belongs to the normal closure of u for some (injective) endomorphism f of the group F , this f is actually an automorphism. Give a particular example of a strong test element.

[Background](#)

*(F18) (V.Shpilrain) An automorphism of a group is said to be *normal* if it leaves invariant every normal subgroup of finite index. A. Lubotzky and A. S.-T. Lue have proved that every normal automorphism of a free group is inner. This yields the following question:

Is there a *single* normal subgroup R of a free group F , so that every automorphism of F that leaves R invariant, is inner?

[Background](#)

*(F19) (M.Wicks) (a) Let F be a non-cyclic free group of rank n , and $P(n, k)$ the number of its primitive elements of length k . What is the growth of $P(n, k)$ as a function of k , with n fixed ?

[Background](#)

(b) The same question for the number of cyclically reduced primitive elements.

(F20) (C.Sims) Is the c -th term of the lower central series of a free group of finite rank the normal closure of basic commutators of weight c ?

[Background](#)

*(F21) (A.Gaglione, D.Spellman) Let F be a non-cyclic free group, and G the Cartesian (unrestricted) product of countably many copies of F . Is the group $G/[G, G]$ torsion-free?

[Background](#)

*(F22) (P.M.Neumann) Let G be a free product amalgamating proper subgroups H and K of A and B , respectively. Suppose that A, B, H, K are free groups of finite ranks. Can G be simple?

[Background](#)

(F23) (A.Olshanskii) Does the free group of rank 2 have an infinite ascending chain of fully invariant subgroups, each being generated (as a fully invariant subgroup) by a single element?

(F24) (A.Miasnikov, V.Remeslennikov) Let G be a free product of two isomorphic free groups of finite ranks amalgamated over a finitely generated subgroup.

*(a) Is the conjugacy problem solvable in G ?

(b) Is there an algorithm to decide if G is free ?

(c) Is there an algorithm to decide if G is hyperbolic?

[Background](#)

(F25) (A.Miasnikov, V.Shpilrain) Let u be an element of a free group F_n , whose length $|u|$ cannot be decreased by any automorphism of F_n . Let $A(u)$ denote the set of elements $\{v \in F_n; |v| = |u|, f(v) = u \text{ for some } f \in \text{Aut}(F_n)\}$.

(a) Is it true that the cardinality of $A(u)$ is bounded by a polynomial function of $|u|$?

*(b) If the free group has rank 2, is it true that the cardinality of $A(u)$ is bounded by $c|u|^2$ for some constant c , which is independent of u ?

[Background](#)

(F26) (M.Bestvina) Let ϕ, ψ be two automorphisms of a free group F_n . Is it true that the intersection of $\text{Fix}(\phi)$ and $\text{Fix}(\psi)$ equals $\text{Fix}(\alpha)$ for some automorphism α of F_n ?

[Background](#)

(F27) (W.Magnus) Let u be an element of a free group F_n . An element r in F_n is called a normal root of u if u belongs to the normal closure of r in the group F_n . Can an element u , which does not belong to the commutator subgroup $[F_n, F_n]$, have infinitely many non-conjugate normal roots ?

[Background](#)

(F28) (S.Sidki) Let S be a subgroup of index 2 in the group F_2 , and let R be an isomorphic copy of S (in F_2). Denote by f an isomorphism between S and R . Is there necessarily a non-trivial subgroup H in S which is invariant under f ? [Background](#)

*(F29) (W. Dicks, E. Ventura) Let H be a subgroup of a free group F_n , and let $r(H)$ denote the rank of H . We call H *inert* if $r(H \cap K)$ is not bigger than $r(K)$ for any subgroup K of F_n . Is every retract of F_n inert? [Background](#)

(F30) (W. Dicks, E. Ventura) Let H be a subgroup of a free group F_n , and let $r(H)$ denote the rank of H . We call H *compressed* if $r(H)$ is not bigger than $r(K)$ for any subgroup K containing H . If H is compressed in F_n , is H necessarily inert? (See the previous problem (F29)). [Background](#)

(F31) (J. Stallings) The equalizer of two homomorphisms $\alpha, \beta: F_n \rightarrow F_m$ is the group $\text{Eq}(\alpha, \beta) = \{x \in F_n : \alpha(x) = \beta(x)\}$. Is it true that if α is injective, then the rank of $\text{Eq}(\alpha, \beta)$ is at most n ? [Background](#)

(F32) (a) An automorphism of a free group F is called an IA-automorphism if it is identical on the Abelianization $F/[F, F]$. Obviously, all IA-automorphisms form a (normal) subgroup $\text{IA}(F)$ of the group $\text{Aut}(F)$. Is the group $\text{IA}(F_n)$ finitely presented for $n > 3$?

*(b) (Yu. Merzlyakov) Is the group $\text{IA}(F_n)$ linear for $n > 2$? [Background](#)

*(F33) (A. Casson) Let f be an automorphism of F_n . Is it true that there is a subgroup K of finite index in F_n , invariant under f , such that every eigenvalue of f_K is equal to a root of 1, where f_K denotes the induced automorphism of $K/[K, K]$? [Background](#)

(F34) (A. Miasnikov, V. Shpilrain) Let F_n be the free group of a finite rank n , with generators $x_1, \dots, x_n$. An element u of F_n is called *positive* if no x_i occurs in u to a negative exponent. An element u is called *potentially positive* if $\alpha(u)$ is positive for some automorphism α of the group F_n . Finally, u is called *stably potentially positive* if it is potentially positive as an element of F_m for some $m \geq n$.

(a) Is the property of being potentially positive algorithmically recognizable?

*(b) Are there stably potentially positive elements of F_n that are not potentially positive?

[Background](#)

(F35) (J. Wiegold) Let R be a characteristic subgroup of a free group $F = F_n$. Can F/R be an infinite simple group? [Background](#)

(F36) Is the group $\text{Out}(F_3)$ linear? [Background](#)

(F37) (V. Bardakov) For any element g of a free group F_n define its *primitive length* as the minimal k such that g is a product of k primitive elements. Is there an algorithm to determine the primitive length of a given element g of F_n ? [Background](#)

(F38) (I. Kapovich, P. Schupp)

(a) Is there an algorithm which, when given two elements u, v of a free group F_n decides whether or not the cyclic length of $f(u)$ equals the cyclic length of $f(v)$ for every automorphism f of the group F_n ?

*(b) Call elements with the property alluded to in part (a) *translation equivalent*, to simplify the language. Is it true that whenever g is translation equivalent to h in F_n and $w(x, y) \in F(x, y)$ is arbitrary, one has $w(g, h)$ translation equivalent to $w(h, g)$ in F_n ?

(c) We say that u is *boundedly translation equivalent* to v if the ratio of the cyclic lengths of $f(u)$ and $f(v)$ is bounded away from 0 and from ∞ . Is there an algorithm which, when given two elements in a finitely generated free group, decides whether or not they are boundedly translation equivalent? [Background](#)

(F39) (O. Bogopolski)

(a) Is there an algorithm which, when given a finitely generated subgroup S of a free group F and an element g of F , decides whether or not there is an automorphism of F that takes g to an element of the subgroup S ?

* (b) The following special case of part (a) is especially attractive: given a finitely generated subgroup S of a free group F , find out whether or not S contains a primitive element of F .

[Background](#)

(F40) (V. Shpilrain)

(a) Let u be an element of a free group F_r . Is it true that there is $v \in F_r$ (that depends on u) that cannot be a subword of any cyclically reduced $f(u)$, where f is an automorphism of F_r ?

(b) A special case of interest is where $u = [x_1, x_2]$, where x_1 and x_2 are elements of the same free generating set of F_r . [Background](#)

(F41) (D. Puder, C. Wu) Suppose u is *not* a primitive element of F_r . Is it true that the exponential growth rate of the automorphic orbit of u is the square root of $2r-1$? [Background](#)

(F42) (V. Dotsenko) Let S_n denote the set of all elements of length n in a free group F_r , and B_n the set of all elements of length $\leq n$.

(a) What is the largest number of elements from S_n that freely generate a subgroup of F_r ?

(b) The same question for elements from B_n .